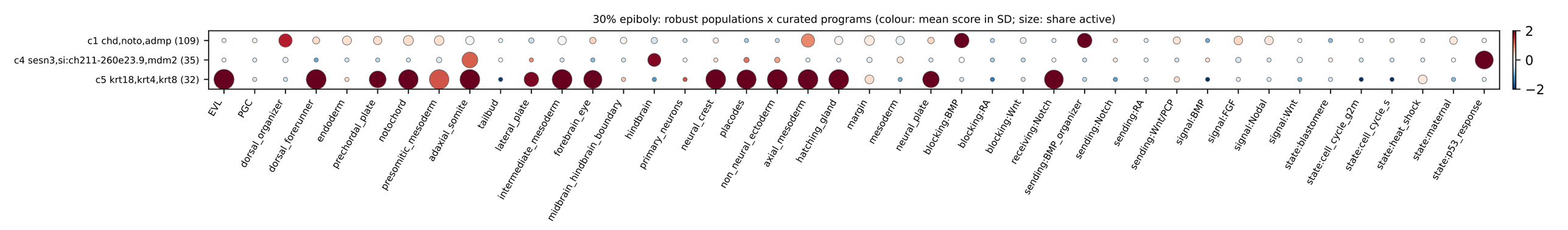
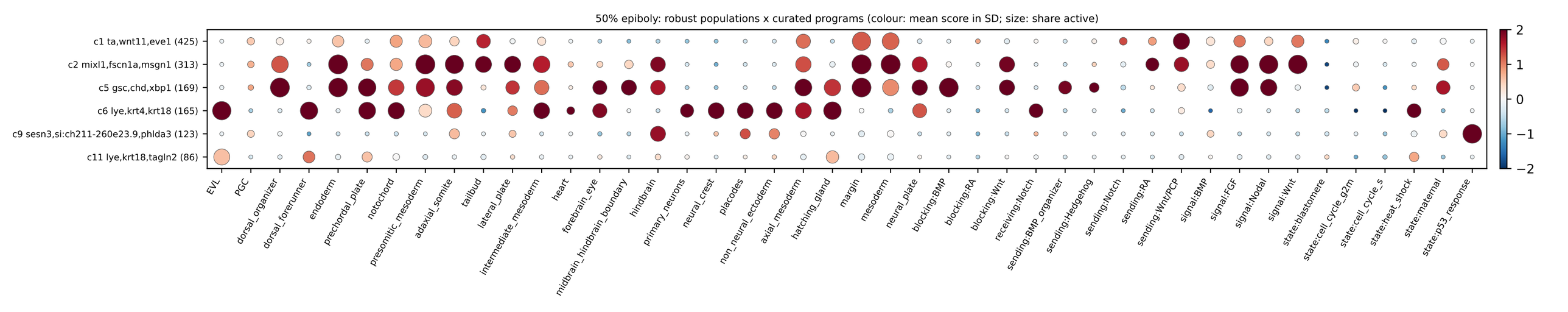
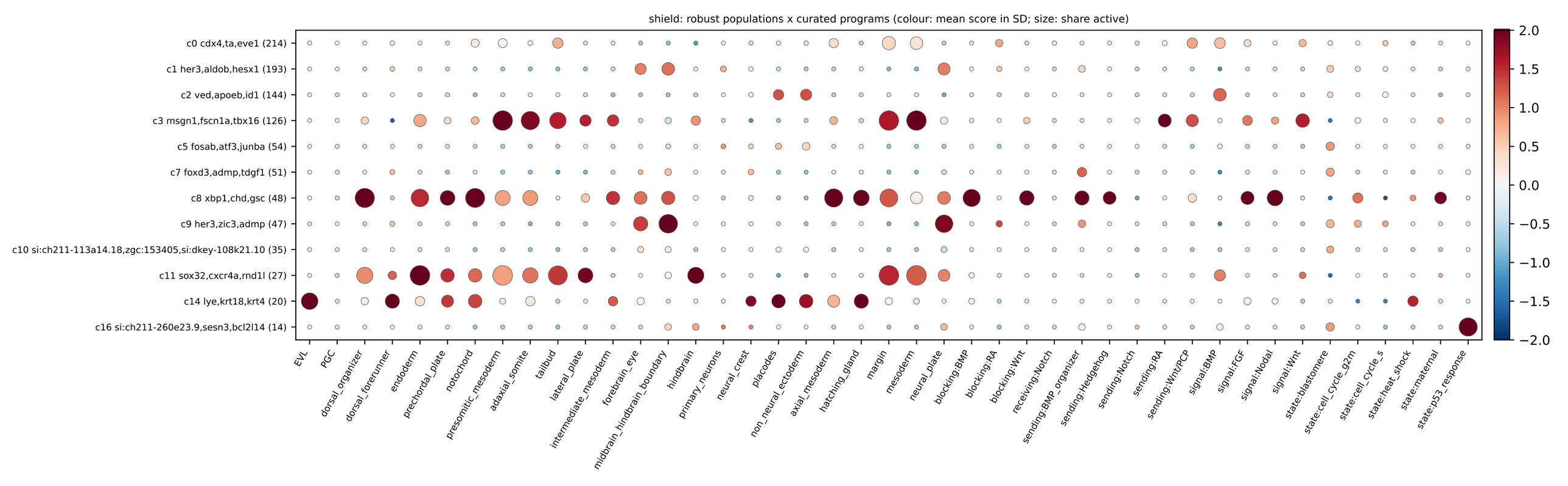
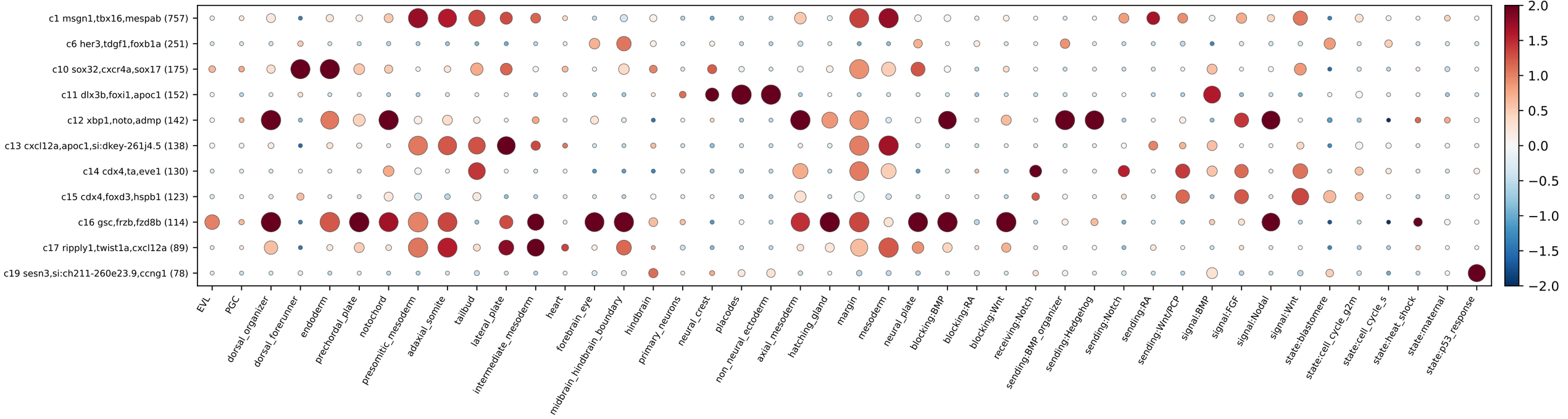
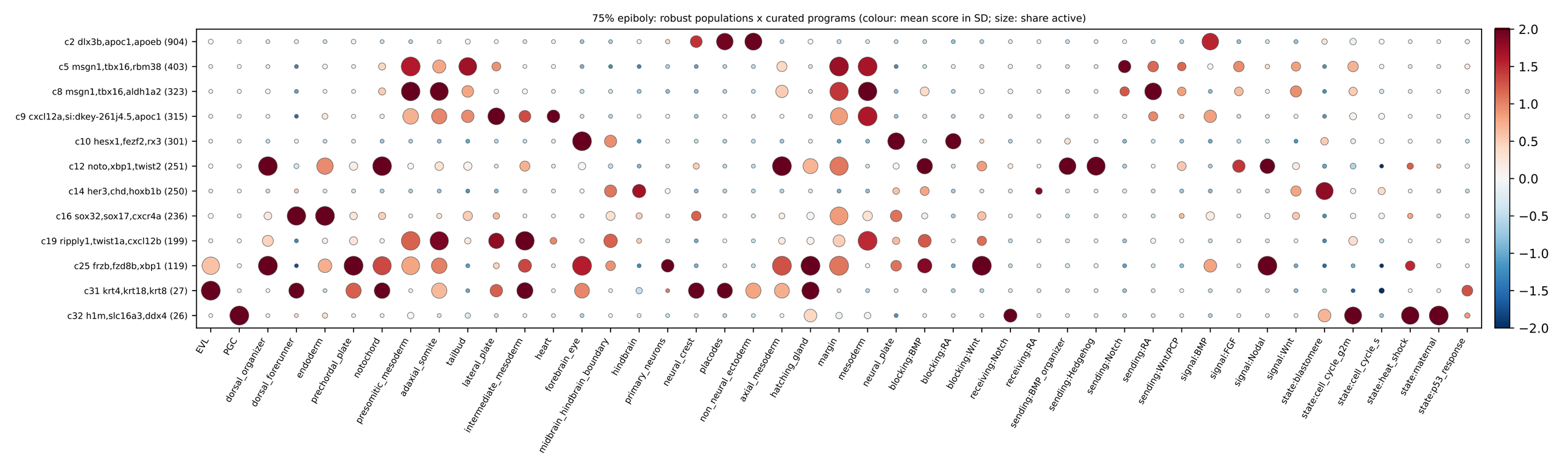

dome: robust populations x curated programs (colour: mean score in SD; size: share active)
c1 chd,gsc,xbp1 (99)
c2 ta,wnt11,mixl1 (98)
c4 lye,krt18,tagln2 (55)
c6 krt18,mid1ip1a,krt4 (39)
c7 zgc:110425,zgc:153405,si:ch211-113a14.12 (39)
EVL
PGC
dorsal_organizer
dorsal_forerunner
endoderm
prechordal_plate
notochord
presomitic_mesoderm
adaxial_somite
tailbud
lateral_plate
intermediate_mesoderm
forebrain_eye
midbrain_hindbrain_boundary
hindbrain
primary_neurons
neural_crest
placodes
non_neural_ectoderm
axial_mesoderm
hatching_gland
margin
mesoderm
neural_plate
blocking:BMP
blocking:RA
blocking:Wnt
receiving:Notch
sending:BMP_organizer
sending:Notch
sending:Wnt/PCP
signal:BMP
signal:FGF
signal:Nodal
signal:Wnt
state:blastomere
state:cell_cycle_g2m
state:cell_cycle_s
state:heat_shock
state:maternal
state:p53_response
2
1
0
−1
−2

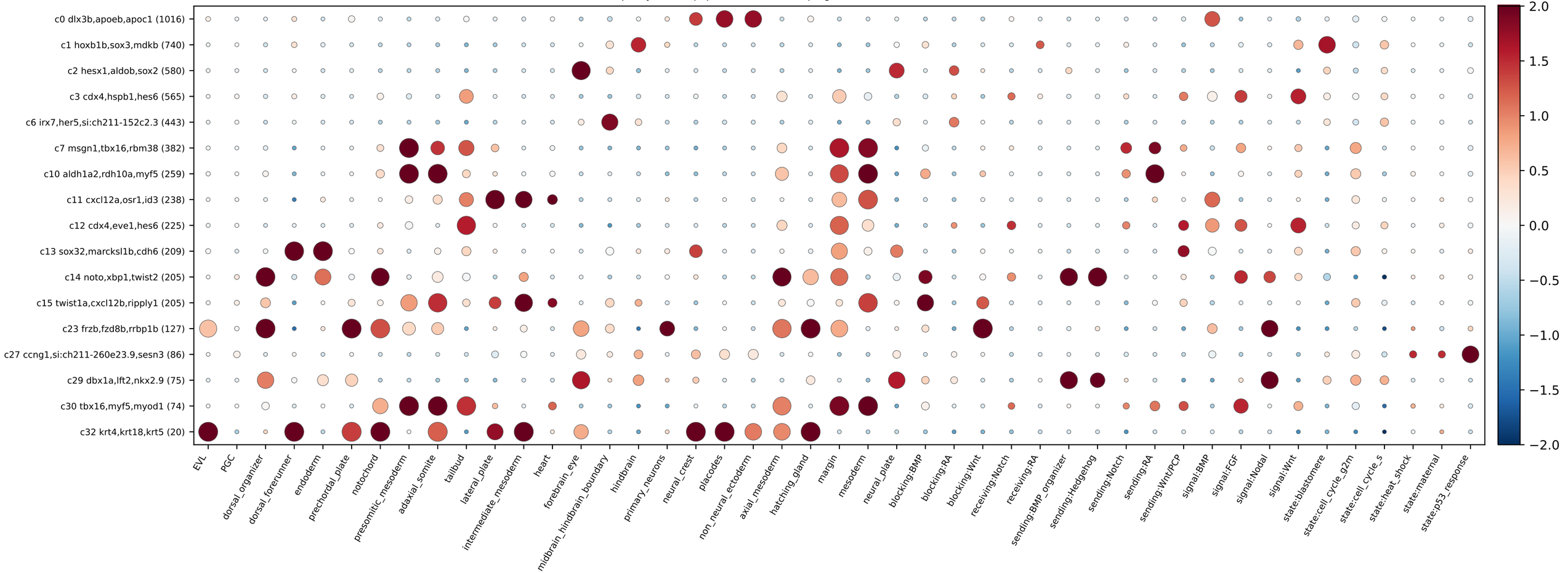

30% epiboly: robust populations x curated programs (colour: mean score in SD; size: share active)
c1 chd,noto,admp (109)
c4 sesn3,si:ch211-260e23.9,mdm2 (35)
c5 krt18,krt4,krt8 (32)
EVL
PGC
dorsal_organizer
dorsal_forerunner
endoderm
prechordal_plate
notochord
presomitic_mesoderm
adaxial_somite
tailbud
lateral_plate
intermediate_mesoderm
forebrain_eye
midbrain_hindbrain_boundary
hindbrain
primary_neurons
neural_crest
placodes
non_neural_ectoderm
axial_mesoderm
hatching_gland
margin
mesoderm
neural_plate
blocking:BMP
blocking:RA
blocking:Wnt
receiving:Notch
sending:BMP_organizer
sending:Notch
sending:RA
sending:Wnt/PCP
signal:BMP
signal:FGF
signal:Nodal
signal:Wnt
state:blastomere
state:cell_cycle_g2m
state:cell_cycle_s
state:heat_shock
state:maternal
state:p53_response
2
0
−2

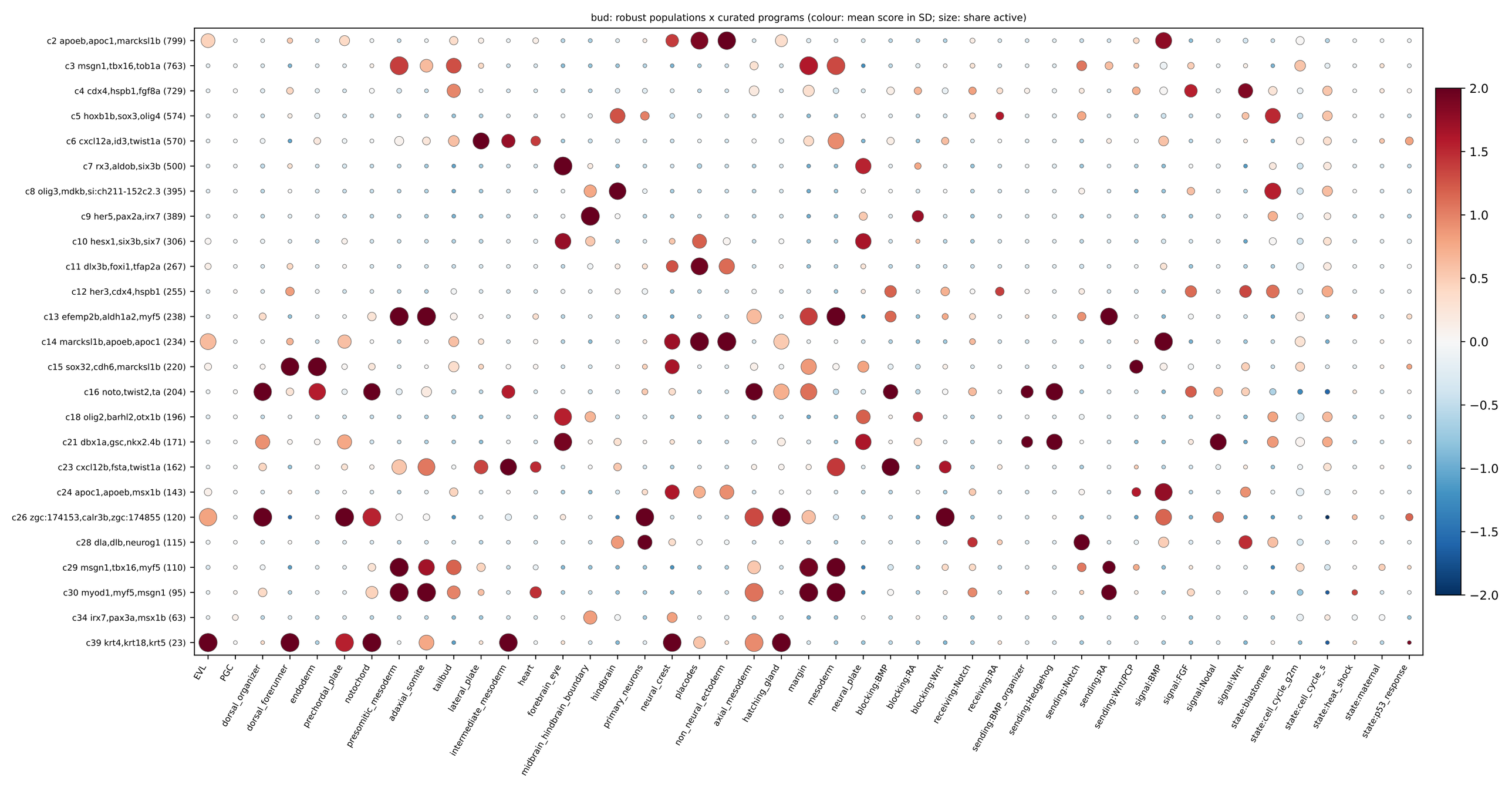

50% epiboly: robust populations x curated programs (colour: mean score in SD; size: share active)
c1 ta,wnt11,eve1 (425)
c2 mixl1,fscn1a,msgn1 (313)
c5 gsc,chd,xbp1 (169)
c6 lye,krt4,krt18 (165)
c9 sesn3,si:ch211-260e23.9,phlda3 (123)
c11 lye,krt18,tagln2 (86)
EVL
PGC
dorsal_organizer
dorsal_forerunner
endoderm
prechordal_plate
notochord
presomitic_mesoderm
adaxial_somite
tailbud
lateral_plate
intermediate_mesoderm
heart
forebrain_eye
midbrain_hindbrain_boundary
hindbrain
primary_neurons
neural_crest
placodes
non_neural_ectoderm
axial_mesoderm
hatching_gland
margin
mesoderm
neural_plate
blocking:BMP
blocking:RA
blocking:Wnt
receiving:Notch
sending:BMP_organizer
sending:Hedgehog
sending:Notch
sending:RA
sending:Wnt/PCP
signal:BMP
signal:FGF
signal:Nodal
signal:Wnt
state:blastomere
state:cell_cycle_g2m
state:cell_cycle_s
state:heat_shock
state:maternal
state:p53_response
2
1
0
−1
−2

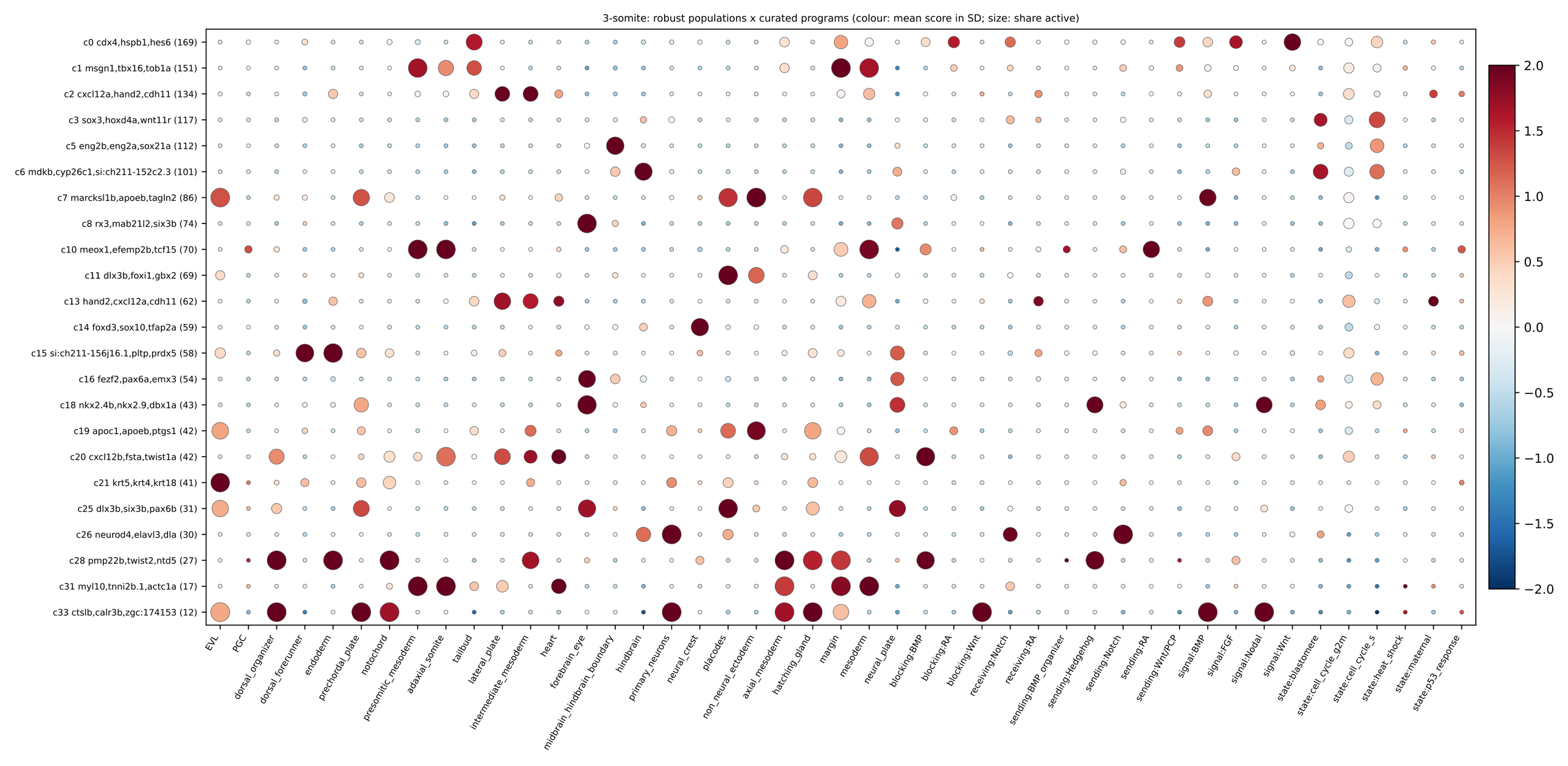

shield: robust populations x curated programs (colour: mean score in SD; size: share active)
c0 cdx4,ta,eve1 (214)
c1 her3,aldob,hesx1 (193)
c2 ved,apoeb,id1 (144)
c3 msgn1,fscn1a,tbx16 (126)
c5 fosab,atf3,junba (54)
c7 foxd3,admp,tdgf1 (51)
c8 xbp1,chd,gsc (48)
c9 her3,zic3,admp (47)
c10 si:ch211-113a14.18,zgc:153405,si:dkey-108k21.10 (35)
c11 sox32,cxcr4a,rnd1l (27)
c14 lye,krt18,krt4 (20)
c16 si:ch211-260e23.9,sesn3,bcl2l14 (14)
EVL
PGC
dorsal_organizer
dorsal_forerunner
endoderm
prechordal_plate
notochord
presomitic_mesoderm
adaxial_somite
tailbud
lateral_plate
intermediate_mesoderm
forebrain_eye
midbrain_hindbrain_boundary
hindbrain
primary_neurons
neural_crest
placodes
non_neural_ectoderm
axial_mesoderm
hatching_gland
margin
mesoderm
neural_plate
blocking:BMP
blocking:RA
blocking:Wnt
receiving:Notch
sending:BMP_organizer
sending:Hedgehog
sending:Notch
sending:RA
sending:Wnt/PCP
signal:BMP
signal:FGF
signal:Nodal
signal:Wnt
state:blastomere
state:cell_cycle_g2m
state:cell_cycle_s
state:heat_shock
state:maternal
state:p53_response
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0

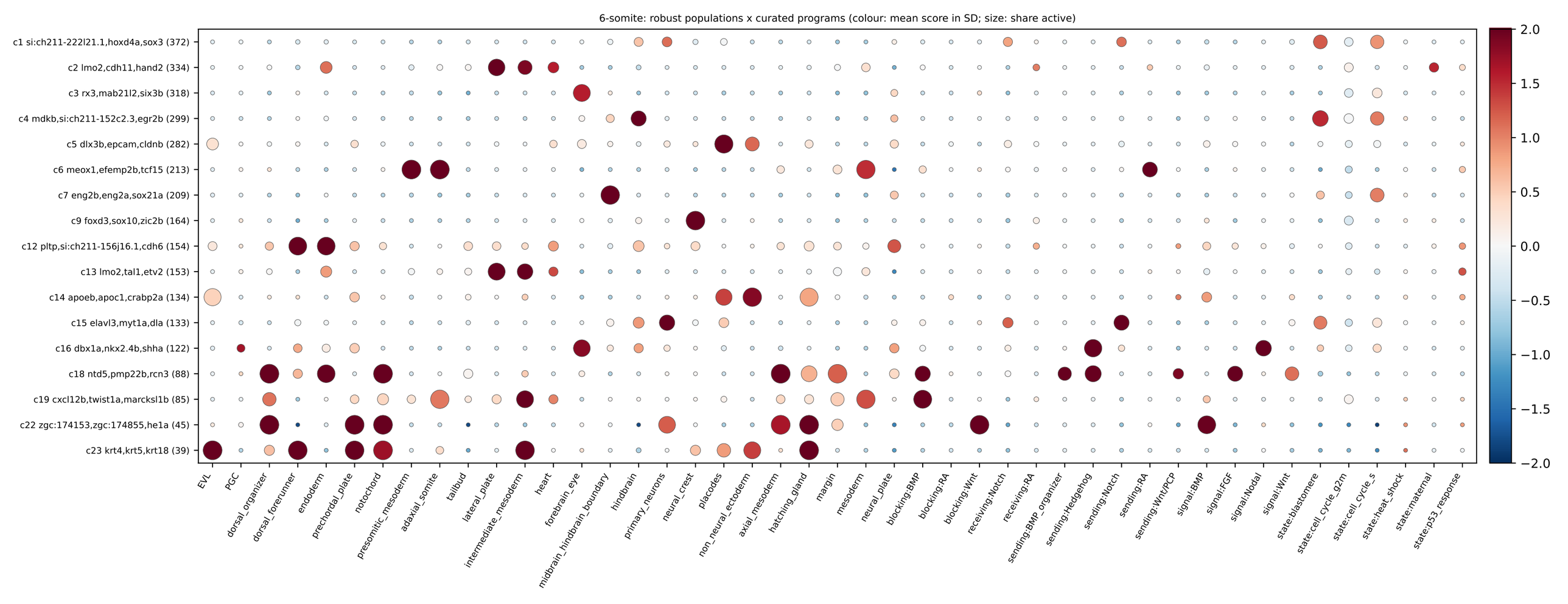

60% epiboly: robust populations x curated programs (colour: mean score in SD; size: share active)
c1 msgn1,tbx16,mespab (757)
c6 her3,tdgf1,foxb1a (251)
c10 sox32,cxcr4a,sox17 (175)
c11 dlx3b,foxi1,apoc1 (152)
c12 xbp1,noto,admp (142)
c13 cxcl12a,apoc1,si:dkey-261j4.5 (138)
c14 cdx4,ta,eve1 (130)
c15 cdx4,foxd3,hspb1 (123)
c16 gsc,frzb,fzd8b (114)
c17 ripply1,twist1a,cxcl12a (89)
c19 sesn3,si:ch211-260e23.9,ccng1 (78)
EVL
PGC
dorsal_organizer
dorsal_forerunner
endoderm
prechordal_plate
notochord
presomitic_mesoderm
adaxial_somite
tailbud
lateral_plate
intermediate_mesoderm
heart
forebrain_eye
midbrain_hindbrain_boundary
hindbrain
primary_neurons
neural_crest
placodes
non_neural_ectoderm
axial_mesoderm
hatching_gland
margin
mesoderm
neural_plate
blocking:BMP
blocking:RA
blocking:Wnt
receiving:Notch
sending:BMP_organizer
sending:Hedgehog
sending:Notch
sending:RA
sending:Wnt/PCP
signal:BMP
signal:FGF
signal:Nodal
signal:Wnt
state:blastomere
state:cell_cycle_g2m
state:cell_cycle_s
state:heat_shock
state:maternal
state:p53_response
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0


75% epiboly: robust populations x curated programs (colour: mean score in SD; size: share active)
c2 dlx3b,apoc1,apoeb (904)
c5 msgn1,tbx16,rbm38 (403)
c8 msgn1,tbx16,aldh1a2 (323)
c9 cxcl12a,si:dkey-261j4.5,apoc1 (315)
c10 hesx1,fezf2,rx3 (301)
c12 noto,xbp1,twist2 (251)
c14 her3,chd,hoxb1b (250)
c16 sox32,sox17,cxcr4a (236)
c19 ripply1,twist1a,cxcl12b (199)
c25 frzb,fzd8b,xbp1 (119)
c31 krt4,krt18,krt8 (27)
c32 h1m,slc16a3,ddx4 (26)
EVL
PGC
dorsal_organizer
dorsal_forerunner
endoderm
prechordal_plate
notochord
presomitic_mesoderm
adaxial_somite
tailbud
lateral_plate
intermediate_mesoderm
heart
forebrain_eye
midbrain_hindbrain_boundary
hindbrain
primary_neurons
neural_crest
placodes
non_neural_ectoderm
axial_mesoderm
hatching_gland
margin
mesoderm
neural_plate
blocking:BMP
blocking:RA
blocking:Wnt
receiving:Notch
receiving:RA
sending:BMP_organizer
sending:Hedgehog
sending:Notch
sending:RA
sending:Wnt/PCP
signal:BMP
signal:FGF
signal:Nodal
signal:Wnt
state:blastomere
state:cell_cycle_g2m
state:cell_cycle_s
state:heat_shock
state:maternal
state:p53_response
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0


90% epiboly: robust populations x curated programs (colour: mean score in SD; size: share active)
c0 dlx3b,apoeb,apoc1 (1016)
c1 hoxb1b,sox3,mdkb (740)
c2 hesx1,aldob,sox2 (580)
c3 cdx4,hspb1,hes6 (565)
c6 irx7,her5,si:ch211-152c2.3 (443)
c7 msgn1,tbx16,rbm38 (382)
c10 aldh1a2,rdh10a,myf5 (259)
c11 cxcl12a,osr1,id3 (238)
c12 cdx4,eve1,hes6 (225)
c13 sox32,marcksl1b,cdh6 (209)
c14 noto,xbp1,twist2 (205)
c15 twist1a,cxcl12b,ripply1 (205)
c23 frzb,fzd8b,rrbp1b (127)
c27 ccng1,si:ch211-260e23.9,sesn3 (86)
c29 dbx1a,lft2,nkx2.9 (75)
c30 tbx16,myf5,myod1 (74)
c32 krt4,krt18,krt5 (20)
EVL
PGC
dorsal_organizer
dorsal_forerunner
endoderm
prechordal_plate
notochord
presomitic_mesoderm
adaxial_somite
tailbud
lateral_plate
intermediate_mesoderm
heart
forebrain_eye
midbrain_hindbrain_boundary
hindbrain
primary_neurons
neural_crest
placodes
non_neural_ectoderm
axial_mesoderm
hatching_gland
margin
mesoderm
neural_plate
blocking:BMP
blocking:RA
blocking:Wnt
receiving:Notch
receiving:RA
sending:BMP_organizer
sending:Hedgehog
sending:Notch
sending:RA
sending:Wnt/PCP
signal:BMP
signal:FGF
signal:Nodal
signal:Wnt
state:blastomere
state:cell_cycle_g2m
state:cell_cycle_s
state:heat_shock
state:maternal
state:p53_response
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0


bud: robust populations x curated programs (colour: mean score in SD; size: share active)
c2 apoeb,apoc1,marcksl1b (799)
c3 msgn1,tbx16,tob1a (763)
c4 cdx4,hspb1,fgf8a (729)
c5 hoxb1b,sox3,olig4 (574)
c6 cxcl12a,id3,twist1a (570)
c7 rx3,aldob,six3b (500)
c8 olig3,mdkb,si:ch211-152c2.3 (395)
c9 her5,pax2a,irx7 (389)
c10 hesx1,six3b,six7 (306)
c11 dlx3b,foxi1,tfap2a (267)
c12 her3,cdx4,hspb1 (255)
c13 efemp2b,aldh1a2,myf5 (238)
c14 marcksl1b,apoeb,apoc1 (234)
c15 sox32,cdh6,marcksl1b (220)
c16 noto,twist2,ta (204)
c18 olig2,barhl2,otx1b (196)
c21 dbx1a,gsc,nkx2.4b (171)
c23 cxcl12b,fsta,twist1a (162)
c24 apoc1,apoeb,msx1b (143)
c26 zgc:174153,calr3b,zgc:174855 (120)
c28 dla,dlb,neurog1 (115)
c29 msgn1,tbx16,myf5 (110)
c30 myod1,myf5,msgn1 (95)
c34 irx7,pax3a,msx1b (63)
c39 krt4,krt18,krt5 (23)
EVL
PGC
dorsal_organizer
dorsal_forerunner
endoderm
prechordal_plate
notochord
presomitic_mesoderm
adaxial_somite
tailbud
lateral_plate
intermediate_mesoderm
heart
forebrain_eye
midbrain_hindbrain_boundary
hindbrain
primary_neurons
neural_crest
placodes
non_neural_ectoderm
axial_mesoderm
hatching_gland
margin
mesoderm
neural_plate
blocking:BMP
blocking:RA
blocking:Wnt
receiving:Notch
receiving:RA
sending:BMP_organizer
sending:Hedgehog
sending:Notch
sending:RA
sending:Wnt/PCP
signal:BMP
signal:FGF
signal:Nodal
signal:Wnt
state:blastomere
state:cell_cycle_g2m
state:cell_cycle_s
state:heat_shock
state:maternal
state:p53_response
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0


3-somite: robust populations x curated programs (colour: mean score in SD; size: share active)
c0 cdx4,hspb1,hes6 (169)
c1 msgn1,tbx16,tob1a (151)
c2 cxcl12a,hand2,cdh11 (134)
c3 sox3,hoxd4a,wnt11r (117)
c5 eng2b,eng2a,sox21a (112)
c6 mdkb,cyp26c1,si:ch211-152c2.3 (101)
c7 marcksl1b,apoeb,tagln2 (86)
c8 rx3,mab21l2,six3b (74)
c10 meox1,efemp2b,tcf15 (70)
c11 dlx3b,foxi1,gbx2 (69)
c13 hand2,cxcl12a,cdh11 (62)
c14 foxd3,sox10,tfap2a (59)
c15 si:ch211-156j16.1,pltp,prdx5 (58)
c16 fezf2,pax6a,emx3 (54)
c18 nkx2.4b,nkx2.9,dbx1a (43)
c19 apoc1,apoeb,ptgs1 (42)
c20 cxcl12b,fsta,twist1a (42)
c21 krt5,krt4,krt18 (41)
c25 dlx3b,six3b,pax6b (31)
c26 neurod4,elavl3,dla (30)
c28 pmp22b,twist2,ntd5 (27)
c31 myl10,tnni2b.1,actc1a (17)
c33 ctslb,calr3b,zgc:174153 (12)
EVL
PGC
dorsal_organizer
dorsal_forerunner
endoderm
prechordal_plate
notochord
presomitic_mesoderm
adaxial_somite
tailbud
lateral_plate
intermediate_mesoderm
heart
forebrain_eye
midbrain_hindbrain_boundary
hindbrain
primary_neurons
neural_crest
placodes
non_neural_ectoderm
axial_mesoderm
hatching_gland
margin
mesoderm
neural_plate
blocking:BMP
blocking:RA
blocking:Wnt
receiving:Notch
receiving:RA
sending:BMP_organizer
sending:Hedgehog
sending:Notch
sending:RA
sending:Wnt/PCP
signal:BMP
signal:FGF
signal:Nodal
signal:Wnt
state:blastomere
state:cell_cycle_g2m
state:cell_cycle_s
state:heat_shock
state:maternal
state:p53_response
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0


6-somite: robust populations x curated programs (colour: mean score in SD; size: share active)
c1 si:ch211-222l21.1,hoxd4a,sox3 (372)
c2 lmo2,cdh11,hand2 (334)
c3 rx3,mab21l2,six3b (318)
c4 mdkb,si:ch211-152c2.3,egr2b (299)
c5 dlx3b,epcam,cldnb (282)
c6 meox1,efemp2b,tcf15 (213)
c7 eng2b,eng2a,sox21a (209)
c9 foxd3,sox10,zic2b (164)
c12 pltp,si:ch211-156j16.1,cdh6 (154)
c13 lmo2,tal1,etv2 (153)
c14 apoeb,apoc1,crabp2a (134)
c15 elavl3,myt1a,dla (133)
c16 dbx1a,nkx2.4b,shha (122)
c18 ntd5,pmp22b,rcn3 (88)
c19 cxcl12b,twist1a,marcksl1b (85)
c22 zgc:174153,zgc:174855,he1a (45)
c23 krt4,krt5,krt18 (39)
EVL
PGC
dorsal_organizer
dorsal_forerunner
endoderm
prechordal_plate
notochord
presomitic_mesoderm
adaxial_somite
tailbud
lateral_plate
intermediate_mesoderm
heart
forebrain_eye
midbrain_hindbrain_boundary
hindbrain
primary_neurons
neural_crest
placodes
non_neural_ectoderm
axial_mesoderm
hatching_gland
margin
mesoderm
neural_plate
blocking:BMP
blocking:RA
blocking:Wnt
receiving:Notch
receiving:RA
sending:BMP_organizer
sending:Hedgehog
sending:Notch
sending:RA
sending:Wnt/PCP
signal:BMP
signal:FGF
signal:Nodal
signal:Wnt
state:blastomere
state:cell_cycle_g2m
state:cell_cycle_s
state:heat_shock
state:maternal
state:p53_response
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0